Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
32	20	Space	48	30	0	64	40	@	80	50	P	96	60	`	112	70	p
33	21	!	49	31	1	65	41	A	81	51	Q	97	61	a	113	71	q
34	22	"	50	32	2	66	42	B	82	52	R	98	62	b	114	72	r
35	23	#	51	33	3	67	43	C	83	53	S	99	63	c	115	73	s
36	24	$	52	34	4	68	44	D	84	54	T	100	64	d	116	74	t
37	25	%	53	35	5	69	45	E	85	55	U	101	65	e	117	75	u
38	26	&	54	36	6	70	46	F	86	56	V	102	66	f	118	76	v
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45	2D	-	61	3D	=	77	4D	M	93	5D	]	109	6D	m	125	7D	}
46	2E	.	62	3E	>	78	4E	N	94	5E	^	110	6E	n	126	7E	~
47	2F	/	63	3F	?	79	4F	O	95	5F	_	111	6F	o			

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Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

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Background

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- Information Management: studies how to **store** and **organize** data

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Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

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  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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Existing Solutions

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Existing Solutions

- **Resource Description Language (RDF)**
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - — No built-in logic engine
- **Labeled Property Graphs (LPGs)**
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - — Not standardized
- **Cyc**
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - — Hard-coded, proprietary rules

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Proposed Solution

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- **Welkin**: an information language

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- **Welkin**: an information language
  - Pronounced: “VEL-KIN”

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- **Welkin**: an information language
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- **Goals:**

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    - Describe improvements *in* Welkin

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General Requirements

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General Requirements

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   - Established notion: **Turing machines**[1]
2. Enable user-defined *meaning*

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- Achieve both through **representations**

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- Centered around *representing* things
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- Track *faithfulness* of representations through **truth**

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- Can be broken down, combined
  - Written with braces $\{a, b, \ldots, z\}$

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- Representations are units
- Each unit has its own **context**

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Information

- Representations

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Information

- Representations $\sim$ **Truth**
- Information

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Information

- Representations ~ **Truth**
- Information ~ **Proofs**
-

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Rules

- **Internal Transitivity:**
- **Contextual Lifting:**

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Rules

- **Internal Transitivity:**
- **Contextual Lifting:**
- **Lattice Laws:**
  - Units as sets

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Turing Completeness

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- Part of Universality
- **Theorem:**

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- **Theorem:** *every unit corresponds to some program, and vice versa.*
  - φ: some mapping between units and programs
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- Later:

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- Later: completeness for **proofs**!

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Encoding

- Characters written in `type-writer font`[1]
- Define US-ASCII: Printable (codes: 32-126) + `EOF` (code: 127)
- **Token:** collection of contiguous characters
  - Simplifies grammar

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[1]Font used: Intel One Mono (https://github.com/intel/intel-one-mono), licensed under OFL 1.1 (https://github.com/intel/intel-one-mono/blob/main/OFL.txt).

US-ASCII

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45	2D	-	59	3B	;	73	49	I	87	57	W	101	65	e	115	73	s			

Table 1: Printable US-ASCII codes and glyphs.

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Building Blocks

- **Invertible Syntax Descriptions:** pairs of functions:[1]

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[1] Rendel & Ostermann (2010). Invertible syntax descriptions. Haskell '10.
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  - Simplifies grammar

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The Welkin Grammar

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Outline of Unambiguity

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